Jingyuan **Ma**

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Profile

I am a third-year undergraduate studying artificial intelligence at the Institute for Interdisciplinary Information Sciences, Tsinghua University. I am currently a research intern in Assistant Professor Yuan Yao's MAI Lab at the College of AI, Tsinghua University, focusing on multimodal machine learning. I am also interested in AI agents, sparse attention and model architectures.

Education

Tsinghua University GPA: 3.984/4.000 Rank: 2/92 — *Beijing, China*

Third-year undergraduate (AI track), Institute for Interdisciplinary Information Sciences — *Present*

Research Experience

Research Intern, MAI Lab, College of AI — *Mar. 2026 – Present*

Supervisor: Assistant Professor Yuan Yao, Tsinghua University.
Research focus: video understanding and multimodal long-context modeling.

Research Intern, Institute for Interdisciplinary Information Sciences — *Sep. 2025 – Jan. 2026*

Supervisor: Yi Wu (then Assistant Professor), Tsinghua University.
Research focus: coding agents and asynchronous reinforcement learning.

Publications & Manuscripts

Replay Before You Reply: Second-Look Prompting for Unified Multimodal Encoding
Under review. https://openreview.net/forum?id=XRNcmbQebj

Budgeted Express-Mesh: Traffic-Aware Link Placement and Deadlock-Free Adaptive Routing
Li Cao and **Jingyuan Ma**.
arXiv preprint, 2026. https://arxiv.org/abs/2609.17057

Selected Projects

From Computation to Information to Accuracy: On the Feasibility of Adaptive Visual Token Pruning
https://github.com/tsjyma/qasparse
Investigates question-adaptive visual token pruning through oracle sparsity labeling, text-only and multimodal sparsity classifiers, and evaluation across seven vision-language benchmarks.

BACH: Bridging Agent and Chord Harmony for Long-form Accompaniment Generation
https://github.com/tsjyma/music-agent
Combines a local chord planner with LLM agents for structure, arrangement, instrument writing, and critique. Generates piano, bass, and drum accompaniment from a LilyPond melody, with score and MIDI rendering.

Open-Source Contributions

MiniCPM Support: ms-swift & LlamaFactory — *2026*

Contributed MiniCPM-V-4.6 model and training support across both frameworks, and MiniCPM-OCR-1.0 model loading and supervised fine-tuning support in ms-swift.
MiniCPM-V-4.6: https://github.com/modelscope/ms-swift/pull/9286 (merged)
MiniCPM-V-4.6: https://github.com/hiyouga/LlamaFactory/pull/10472 (merged)
MiniCPM-OCR-1.0: https://github.com/modelscope/ms-swift/pull/10414 (merged)

AReaL — *Merged PR #480, 2025*

Implemented M2PO algorithm support, with an example configuration and documentation.
https://github.com/areal-project/AReaL/pull/480

Leadership & Service

Institute for Interdisciplinary Information Sciences — *Tsinghua University*

Vice President, Science and Technology Association — *Oct. 2026 – Present*

Selected Honors & Awards

2026 **Jiang Nanxiang Scholarship**
2025 **National Scholarship**
2025 **Grand Prize (Rank 1),** Tsinghua University Freshman Mathematics and Physics Competition
2023 **Silver Medal,** 40th Chinese Physics Olympiad (CPhO)